



Fig. 6: Typical laser-guided bomb integrated with laser seeker head

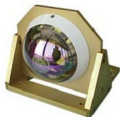


Fig. 7: LADAR sensor



Fig. 8: Laser power/energy meter

In addition, the operational wavelength band may also vary from one application to another. More complex laser warning systems indicate the type of laser threat, the wavelength and the direction of arrival of the threat. A typical high-end laser warning sensor is capable of characterising threats from laser rangefinders,

laser target designators and laser beam riders, and these are available for mounting on helicopters, main battle tanks and light-armoured vehicles.

While one of the most common military applications of a laser warning sensor is as a sub-system in an integrated EOCM system for armoured fighting platforms, there are many other application scenarios where laser sensors are deployed.

Emerging trends indicate the use of an array of laser sensors interfaced with a high-energy laser to protect critical and strategic assets such as aircraft shelters, ammunition depots, strategic buildings, naval vessels and so on from laser-guided munitions attack. Arrays of laser sensors in this case detect laser threat and decode its parameters. These parameters are used to control the operation of a high-energy laser source. The laser source in turn illuminates a dummy target to misguide the incoming laser-guided munitions towards the dummy target.

Laser-guided munitions use a laser sensor called laser seeker, which is also

a kind of position sensor. Laser seeker is the heart of the guidance system of a laser-guided weapon such as laser-guided bomb or missile.

Fig. 6 shows the laser-guided bomb integrated with laser seeker head. A typical laser seeker employs a quadrant sensor for determination of direction of arrival of laser radiation scattered off the intended target when illuminated by a laser target designator. The laser target designator and the laser seeker work in harmony. Both operate on the same pulse repetition frequency (PRF) code, which allows the bomb to home on to the source of laser scatter.

LADAR sensor (Fig. 7) is one of the most contemporary forms of laser seekers usually used for ultra-high precision hitting of strategic targets. It is mainly used in conjunction with other guidance systems on strategic payloads for intended target discrimination from advanced decoys and aim point selection. It is also well suited for combat identification, navigation of autonomous vehicles and topography.

LADAR is also suitable in finding targets hidden by camouflage nets and foliage. LADAR seeker can detect and identify specific features of the target with very high definition up to the resolution of a few centimetres from the distance of a few kilometres.

LADAR sensors are usually employed on loitering systems that look at the target from different angles, verify target's identity and select the best attack position for desired results. The sensor in essence generates a 3D image of the intended target. The 3D image is compared with various 3D templates stored in weapon's memory before the mission and it facilitates identification of target and selection of aim point.

Test and evaluation of lasers

Measurement of laser power, energy, pulse width, etc is yet another area that is electronic or more precisely opto-electronic in nature. Today, we have all

kinds of lasers producing CW or pulsed or Q-switched pulsed laser outputs. While we would be mainly interested in the output power in the CW lasers such as gas lasers, it is the energy and the pulse width that would be of interest in case of pulsed lasers. In case of Q-switched pulsed lasers, such as solid-state lasers, we would like to measure energy per pulse, average power and also the peak power. Equipment that are capable of measuring one or more of these parameters are commercially available. Fig. 8 shows one such commercial meter capable of measuring laser power and pulse energy over a wide range when used in conjunction with suitable sensor heads.

While commercially available test and measurement equipment can be used for carrying measurement of important laser parameters and therefore be useful to the engineers responsible for maintaining military laser equipment, there are cases where dedicated test systems are needed to perform health checks. Sometimes, these quick health checks, also called serviceability checks, are needed to be performed on the systems integrated on the platform.

The point that we are trying to drive home is that while talking or writing about laser electronics, it would be far from being justified to confine the discussion to merely power supplies for different types of lasers, because then we would probably be covering not more than 30 per cent of the electronics that concerns contemporary lasers and laser-based systems.

In the present article, we have presented different areas of usage of laser and opto-electronics systems in defence, and we have briefly touched upon the role of electronics in each of those areas. Details will be presented in the following parts of the article.

To be continued

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